

would be a combination of lighter materials (like carbon fibre composites) for larger cobots, sensing (real-time vision, very sensitive haptic feedback etc) for avoiding unintended contact or providing intentional engagement with humans, and intelligence (reinforcement learning etc) to learn the context of operation to minimise human attention and interventions,” sums up Jagannath Raju, Chief Technology Officer, Systemantics India Pvt Ltd.

See, what I can do for you!

“The introduction of cobots has opened up automation possibilities in almost all industries—manufacturing and service. For example, we have had discussions with companies building automated kiosks and floor cleaning robots. We are also discussing with companies providing automation solutions with cobots for warehouse solutions, greenhouse automation, construction field, and so on,” says Suraj.

Industries like food, plastics, electronics, pharmaceuticals, metal and automotive have been using cobots for several tasks like pick-and-place, palletisation, and warehousing. Now, cobots are seen in more people-facing roles like cleaning, healthcare, and food service. Let us take a look at some of these new applications, which are emerging on top of the already established industrial uses.

Cleaning the nooks and corners.

Yes, the little bots that clean floors are also cobots, because they move around with people and pets in the house. That said, cobots are also being used for cleaning on a more industrial and commercial level. Cobots integrated with floor-cleaning mobile robots help to access the nooks and corners. Many airports, including London, Pittsburgh, and Hong Kong, have deployed cobots for floor clean-



Multiple views of Systemantics' ASYSTR 3C cobot (Courtesy: Systemantics)

ing, so that the humans can focus on sanitising kiosk screens, cleaning workstations, and so on.

Pune based Peppermint Robotics' robotic scrubber dryer is an intelligent and connected robot, replete with eleven levels of safety, four different types of sensors and cameras, dynamic pressure for different types of floors, and a dashboard for real-time reports. It cleans most areas by itself, but if you want to guide it into a difficult corner to clean, you can do so with a remote or by asking it to 'follow' you!

Gary is a household chores cobot developed by Unlimited Robotics, Israel. It can take out the laundry, work with a dishwasher, water your plants, clean the floor and windows, press your clothes, and serve snacks. It is open for pre-booking now. That sounds pretty much like Rosey of the Jetsons' fame.

Professor C. Amarnath, known as the Father of Robotics in India, has mentored several students and their startups. He feels scrubbing of vessels, basins, and toilets is also

a potential task for cobots—restaurants, hospitals, they all need it. It requires a different kind of arm movement but is definitely something that cobot makers should explore.

Assisting in health-care—from tests to talks.

Cobots are slowly being adopted by the healthcare industry for varied tasks, from sample sorting and patient monitoring to elderly care and surgical assistance.

Testing. In labs, cobots help with sample sorting as well as testing. A couple of years ago, ABB demonstrated a mobile lab concept at the Texas Medical Center in Houston. Around 2017, the Copenhagen University Hospital lab was burdened by over 3000 tests per day. Two UR5 cobots were deployed to assist the lab technicians. Even though the workload went up by 20% thereafter, the lab was able to get 90% of the tests done within an hour. Cobots like the Dobot M1 Pro four-axis selective compliance articulated robot arm (SCARA) can pipet and mix chemical liquids for lab testing. During Covid times, cobots assisted several labs to handle the burgeoning load of PCR tests.

Disinfection. In those trying times, many hospitals also used cobots to clean and disinfect beds and rooms, thereby protecting their staff. The eXtremeDisinfection roBOT (XDBOT) developed by the Nanyang Technology University in Singapore comprises a UR5 cobot equipped with an electro-

Is there a place for hobbyists here?

Many innovations at the grassroots level come from hobbyists. And there are quite a few very enthusiastic hobbyists in the robotics space too. So, what is the feasibility of them developing innovative cobots? Here is what Venkatesh Gurappa, mentor to many a roboticist, has to say:

“It is difficult for a hobbyist to develop a cobot, due to constraints like unavailability of components and absence of sufficient open-source alternatives.

“Most of the usable precision components are proprietary and expensive, if at all it is possible to import them, considering the many import-export restrictions on such components. The critical aspect of safety depends on the precision and quality of the moving components and the accuracy of the sensors. The costs are relatively high due to this. The idea may be more feasible if the individual components are indigenously-developed and manufactured in India, and has a greater potential too.

“If hobbyists do manage to build one, getting it to the market is not a problem since the industry itself is looking out for such systems.”